



US012406865B2

(12) **United States Patent**
Babayan et al.

(10) **Patent No.:** **US 12,406,865 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SUBSTRATE CARRIER WITH ARRAY OF
INDEPENDENTLY CONTROLLABLE
HEATER ELEMENTS**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **Applied Materials, Inc.**, Santa Clara,
CA (US)

4,818,327 A 4/1989 Davis et al.
5,610,650 A 3/1997 Itoh
(Continued)

(72) Inventors: **Steven E. Babayan**, Los Altos, CA
(US); **Phillip Criminale**, Livermore,
CA (US); **Zhiqiang Guo**, Sunnyvale,
CA (US)

FOREIGN PATENT DOCUMENTS

CN 103854947 A 6/2014
JP 2014-525664 9/2014

(Continued)

(73) Assignee: **Applied Materials, Inc.**, Santa Clara,
CA (US)

OTHER PUBLICATIONS

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1098 days.

Official Action from Chinese Patent Application No. 201780050548.9
dated Nov. 11, 2002, 11 pgs.

(Continued)

(21) Appl. No.: **17/365,931**

(22) Filed: **Jul. 1, 2021**

Primary Examiner — Jimmy Chou

(74) *Attorney, Agent, or Firm* — Schwabe, Williamson &
Wyatt, P.C.

(65) **Prior Publication Data**

US 2021/0335631 A1 Oct. 28, 2021

Related U.S. Application Data

(62) Division of application No. 15/241,326, filed on Aug.
19, 2016, now abandoned.

(51) **Int. Cl.**

H01L 21/67 (2006.01)

H05B 1/02 (2006.01)

(52) **U.S. Cl.**

CPC .. **H01L 21/67103** (2013.01); **H01L 21/67069**
(2013.01); **H01L 21/67248** (2013.01); **H05B**
1/0233 (2013.01); **H05B 2203/035** (2013.01)

(58) **Field of Classification Search**

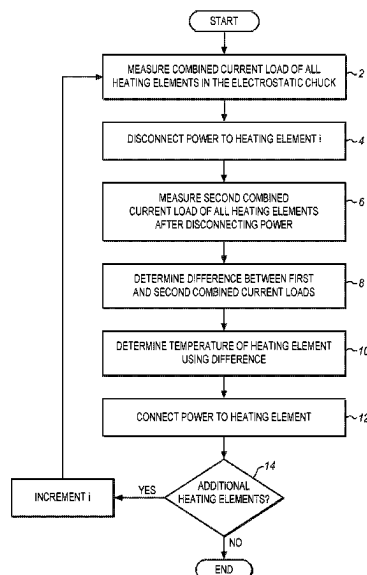
CPC H01L 21/67103; H01L 21/67069; H01L
21/67248; H05B 1/0233; H05B 2203/035

(Continued)

(57) **ABSTRACT**

A substrate carrier is described with an array of indepen-
dently controllable heater elements. In one example an
apparatus includes a substrate carrier to carry a substrate for
processing, a plurality of resistive heating elements in the
carrier to heat the substrate by heating the carrier, a power
supply to supply power to the heating elements, a power
controller to provide a control signal, the control signal to
control an amount of current applied to each of the heating
elements, and a plurality of power interfaces in the carrier
each coupled to a heating element to receive the power from
the power supply and the control signal from the controller
and to modulate the power applied to a respective coupled
heating element in response to the control signal.

6 Claims, 8 Drawing Sheets



(58) **Field of Classification Search**

USPC 219/486
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,846,073 A 12/1998 Weaver
6,403,491 B1 * 6/2002 Liu H01L 21/76802
257/E21.507
6,469,283 B1 * 10/2002 Burkhart H01L 21/67103
219/486
6,496,749 B1 * 12/2002 Yamaguchi H01L 21/67248
219/501
6,580,059 B1 * 6/2003 Kanno H01L 21/67115
219/486
7,612,311 B2 11/2009 Norton et al.
9,029,740 B2 * 5/2015 Aguilar H05K 3/3494
219/486
9,307,578 B2 4/2016 Pease
11,823,874 B2 * 11/2023 Galstyan H01J 37/32174
12,094,695 B2 * 9/2024 Ni H01L 21/6833
2004/0081439 A1 4/2004 Kholodenko et al.
2006/0037701 A1 * 2/2006 Koshiishi H01L 21/31144
257/E21.252
2008/0017104 A1 1/2008 Matyushkin et al.
2008/0116180 A1 5/2008 Norton et al.
2008/0197780 A1 * 8/2008 Yamazawa H01L 21/67069
315/111.21
2009/0014323 A1 * 1/2009 Yendler H01J 37/20
204/298.33
2012/0048467 A1 3/2012 Mahadeswaraswamy et al.
2012/0097661 A1 4/2012 Singh
2012/0111500 A1 * 5/2012 Nagayama H01L 21/6831
118/723 R
2012/0219921 A1 8/2012 Shibagaki et al.
2012/0221138 A1 8/2012 Hong
2013/0098895 A1 4/2013 Swanson et al.
2013/0161305 A1 * 6/2013 Ptasienski H01L 21/67109
219/508
2013/0220989 A1 8/2013 Pease et al.
2013/0270250 A1 * 10/2013 Pease H05B 1/0233
219/443.1
2013/0277339 A1 * 10/2013 Willwerth H01J 37/32724
219/121.43
2014/0048529 A1 * 2/2014 Pease H05B 1/0202
219/494
2014/0120732 A1 * 5/2014 Matsumoto H01L 21/31116
438/723
2014/0144876 A1 * 5/2014 Nakagawa H01J 37/32091
216/51
2014/0154819 A1 * 6/2014 Gaff H01L 22/14
156/345.52
2014/0206108 A1 7/2014 Kiyama
2014/0263274 A1 9/2014 Singh
2015/0116689 A1 * 4/2015 Lee B23K 1/0016
355/72
2015/0132863 A1 * 5/2015 Oohashi H01J 37/32935
438/5
2015/0155193 A1 * 6/2015 Hsu H01L 21/6833
156/345.24
2015/0219499 A1 8/2015 Waldmann et al.
2015/0226611 A1 8/2015 Busche et al.
2015/0228513 A1 * 8/2015 Parkhe H01L 21/6831
219/448.12
2015/0262793 A1 9/2015 Okunishi et al.

2015/0332942 A1 11/2015 Peh et al.
2016/0027678 A1 * 1/2016 Parkhe H01L 21/6833
219/536

2016/0035544 A1 2/2016 Lubomirsky et al.
2016/0104605 A1 4/2016 Hiroki
2016/0149733 A1 * 5/2016 Criminale H04L 25/4902
375/238

2016/0198524 A1 7/2016 Pease et al.
2016/0240366 A1 8/2016 Engelhardt et al.
2016/0370788 A1 12/2016 Bailey, III et al.
2016/0370795 A1 12/2016 Musselman et al.
2016/0370796 A1 12/2016 Musselman et al.
2016/0372352 A1 12/2016 Wu et al.
2016/0372355 A1 12/2016 Zhang et al.
2017/0140957 A1 5/2017 Kitagawa

FOREIGN PATENT DOCUMENTS

JP 2015-084350 4/2015
JP 2015-216254 12/2015
JP 2014-112672 6/2020
KR 10-2014-0070494 6/2014
KR 10-2016-0078332 7/2016
KR 20160078332 A * 7/2016 H01L 21/67
WO WO 2015/119734 8/2015

OTHER PUBLICATIONS

Official Letter from Taiwan Patent Application No. 106127525 dated Aug. 9, 2021, 11 pgs.
International Search Report and Written Opinion from PCT/US2017/045988 mailed Nov. 17, 2017, 10 pgs.
International Preliminary Report on Patentability from PCT/US2017/045988 mailed Feb. 28, 2019, 7 pgs.
Non-Final Office Action from U.S. Appl. No. 15/241,326 dated Jan. 10, 2019, 26 pgs.
Final Office Action from U.S. Appl. No. 15/241,326 dated Apr. 25, 2019, 17 pgs.
Non-Final Office Action from U.S. Appl. No. 15/241,326 dated Jul. 29, 2019, 18 pgs.
Final Office Action from U.S. Appl. No. 15/241,326 dated Dec. 12, 2019, 20 pgs.
Non-Final Office Action from U.S. Appl. No. 15/241,326 dated Mar. 19, 2020, 19 pgs.
Notice to File a Response (Preliminary Rejection) from Korean Patent Application No. 10-2019-7007647 dated May 28, 2020, 7 pgs.
Final Office Action from U.S. Appl. No. 15/241,326 dated Jun. 26, 2020, 20 pgs.
Notice of Reasons for Rejection from Japanese Patent Application No. 2019-509505 dated Jun. 30, 2020, 17 pgs.
Non-Final Office Action from U.S. Appl. No. 15/241,326 dated Oct. 16, 2020, 23 pgs.
Notice of Final Rejection from Korean Patent Application No. 10-2019-7007647 dated Nov. 23, 2020, 7 pgs.
Official Letter from Taiwan Patent Application No. 106127525 dated Jan. 14, 2021, 12 pgs.
Preliminary Rejection from Korean Patent Application No. 10-2019-7007647 dated Jan. 26, 2021, 8 pgs.
Final Office Action from U.S. Appl. No. 15/241,326 dated Apr. 1, 2021, 27 pgs.
Notice of Reasons for Rejection from Japanese Patent Application No. 2019-509505 dated Apr. 12, 2021, 6 pgs.

* cited by examiner

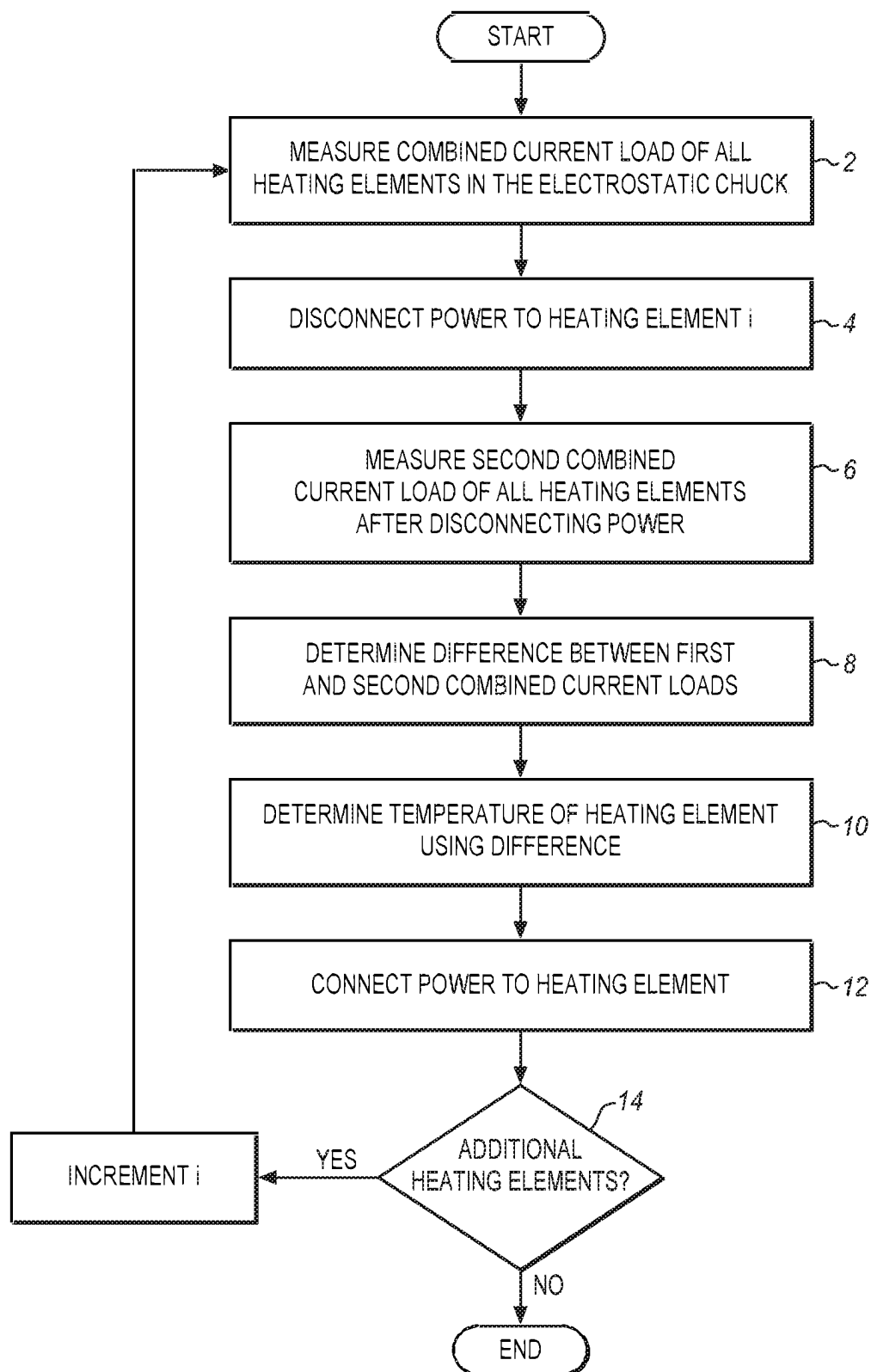
FIG. 1

FIG. 2

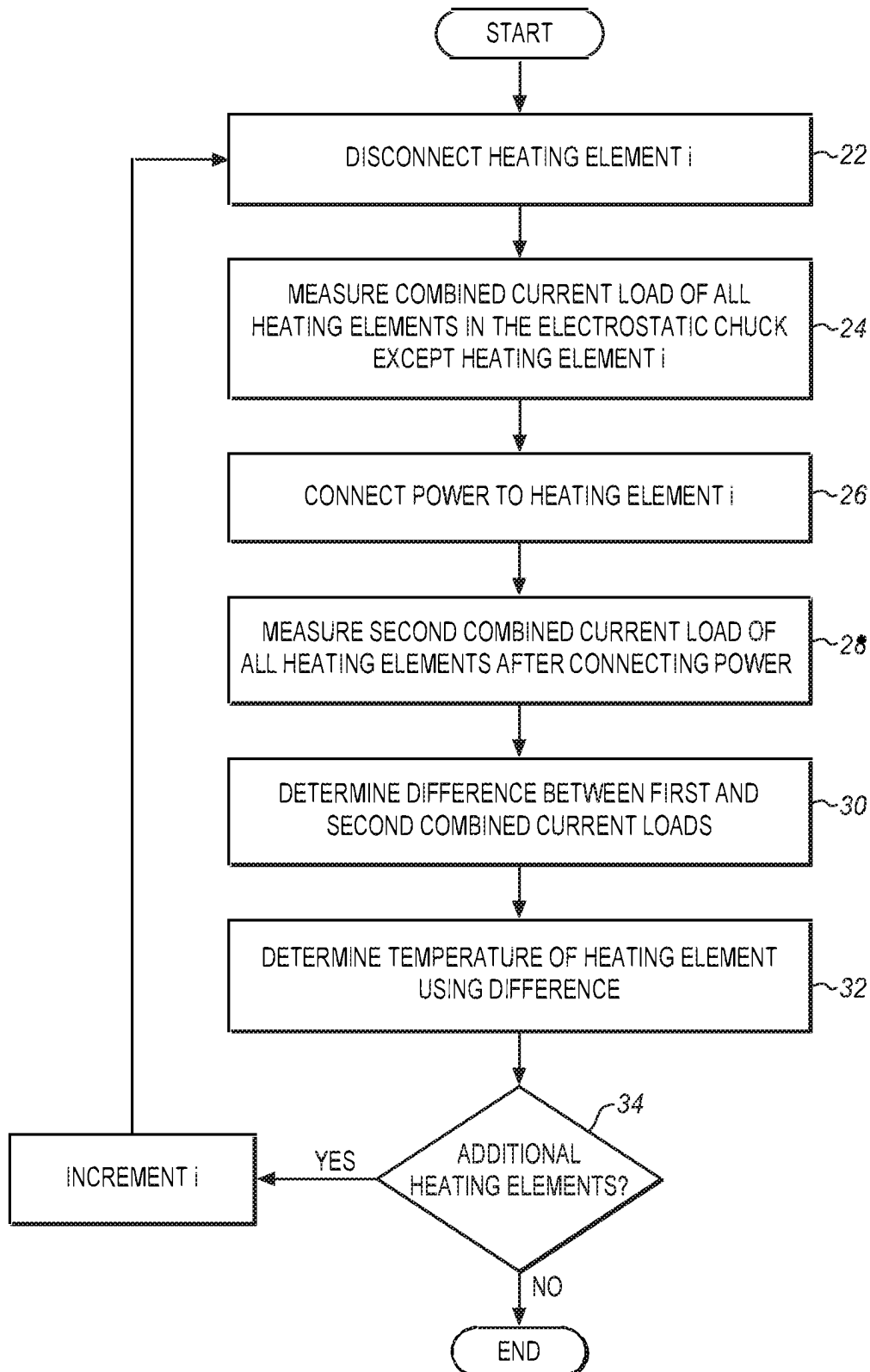


FIG. 3

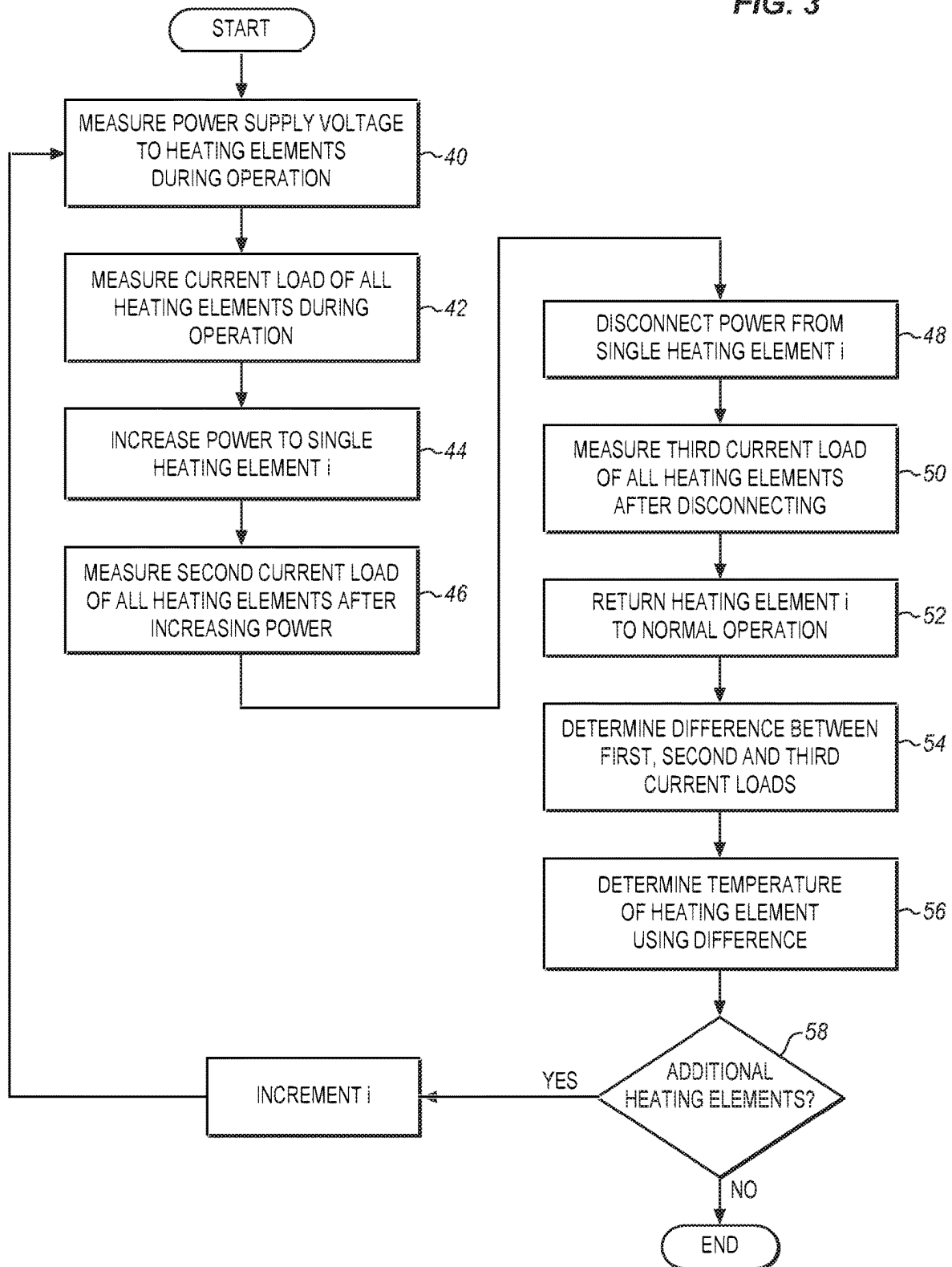


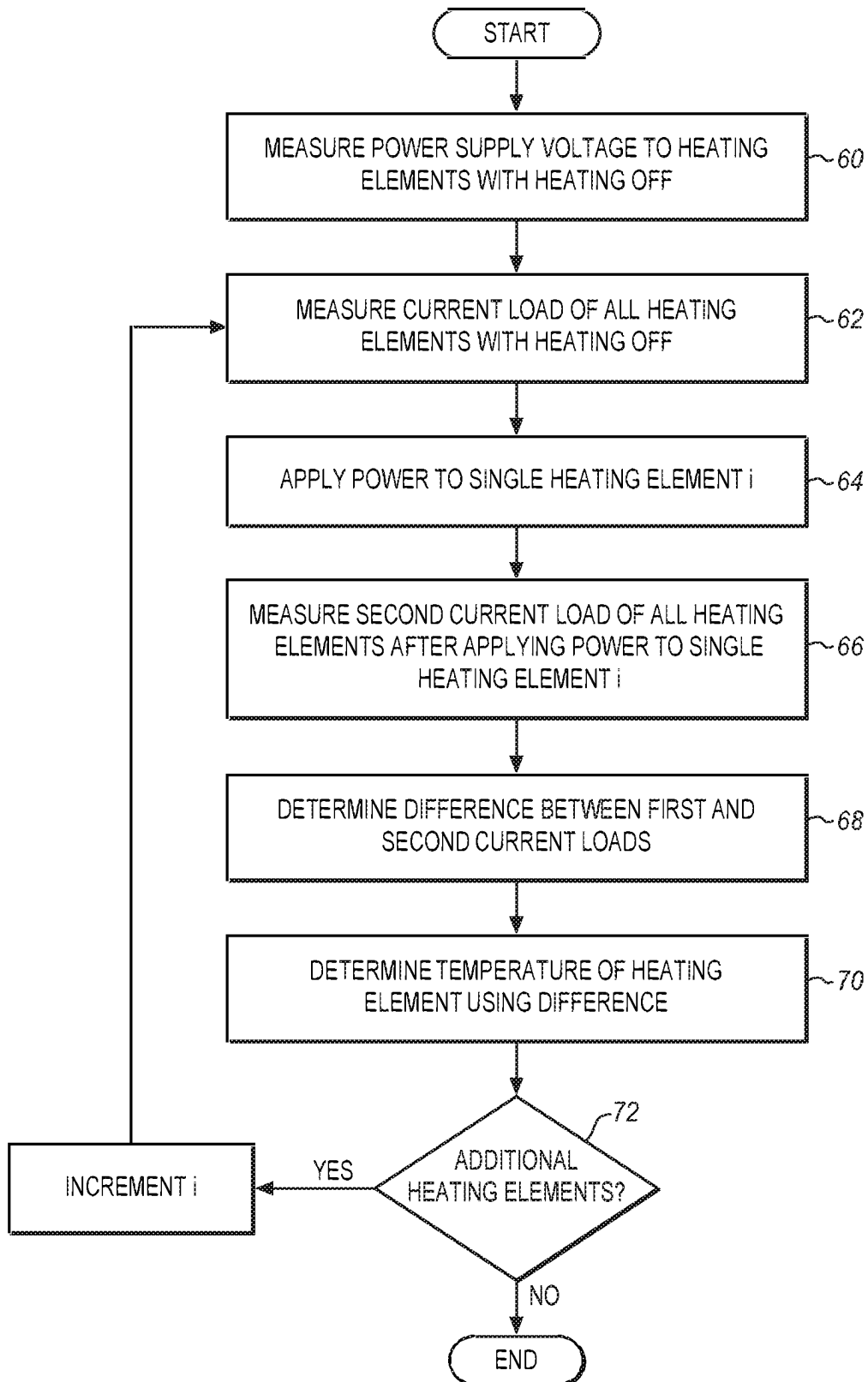
FIG. 4

FIG. 5

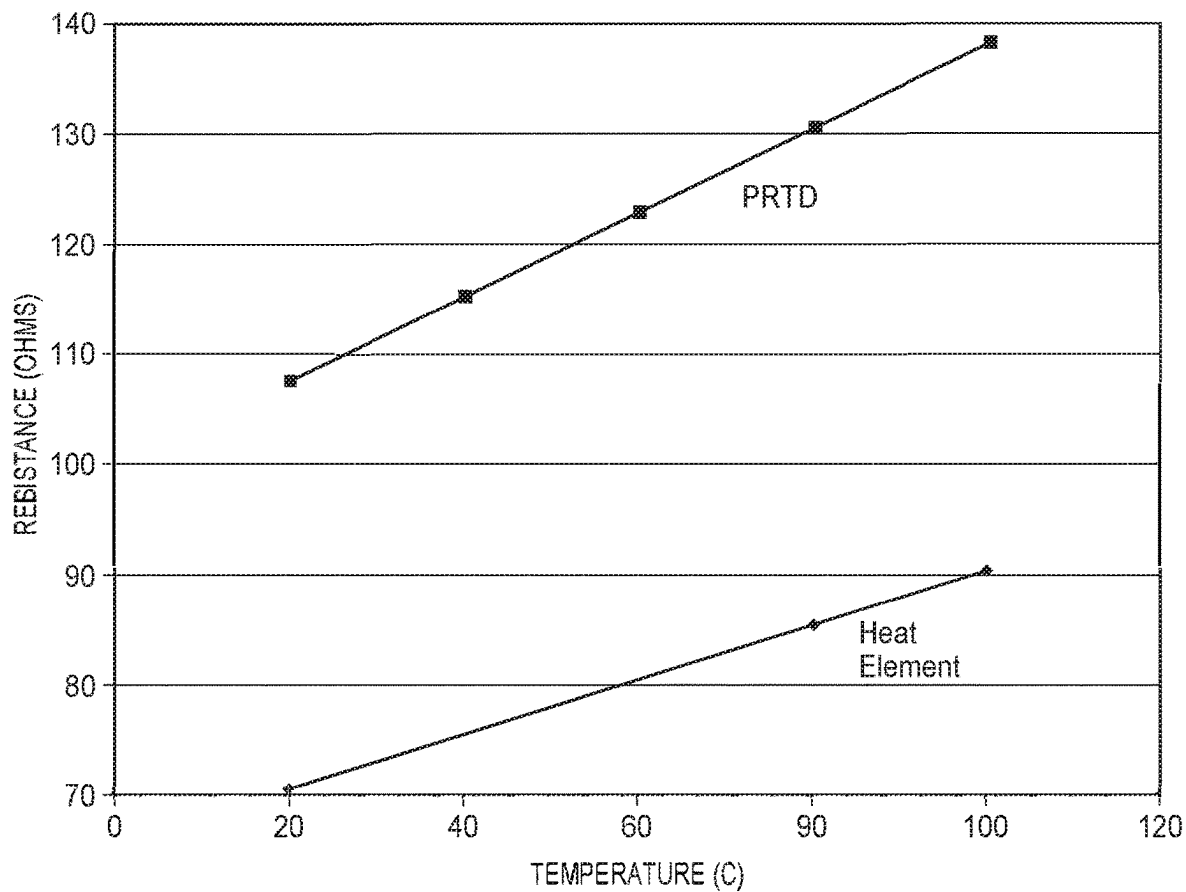


FIG. 6

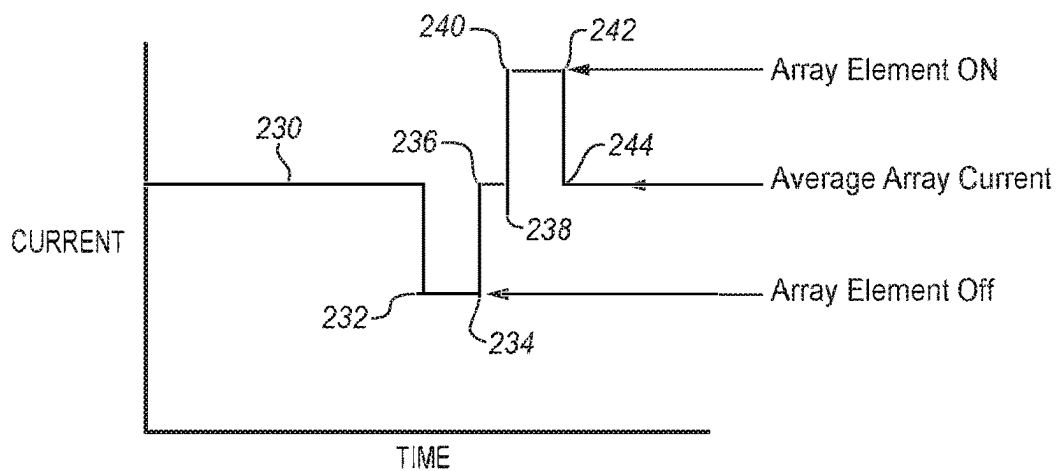


FIG. 7

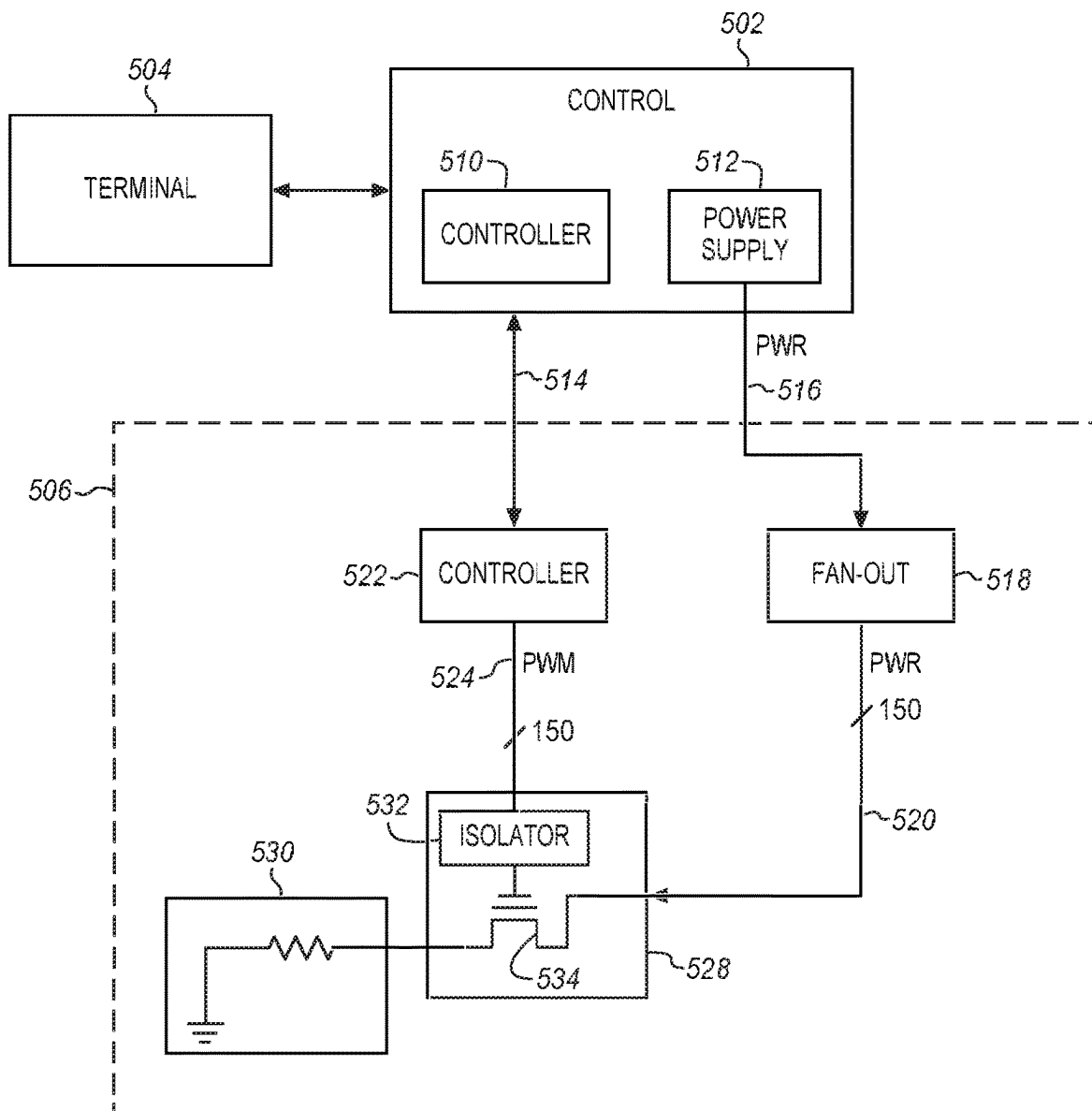


FIG. 8

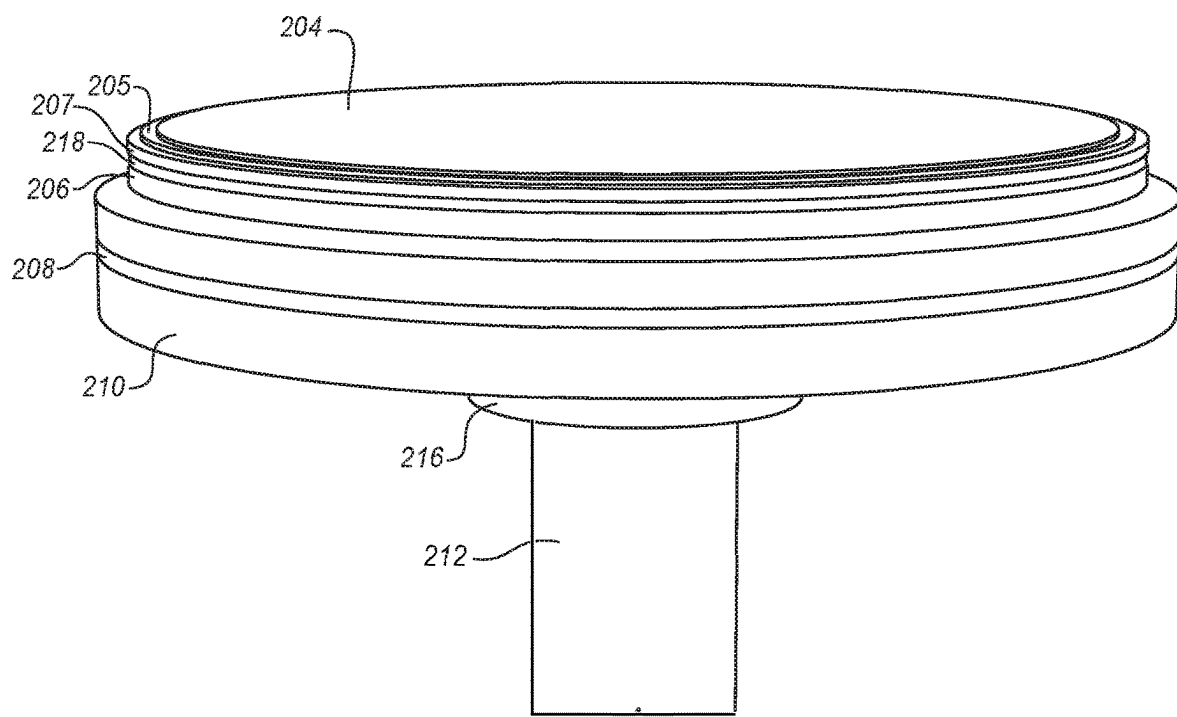
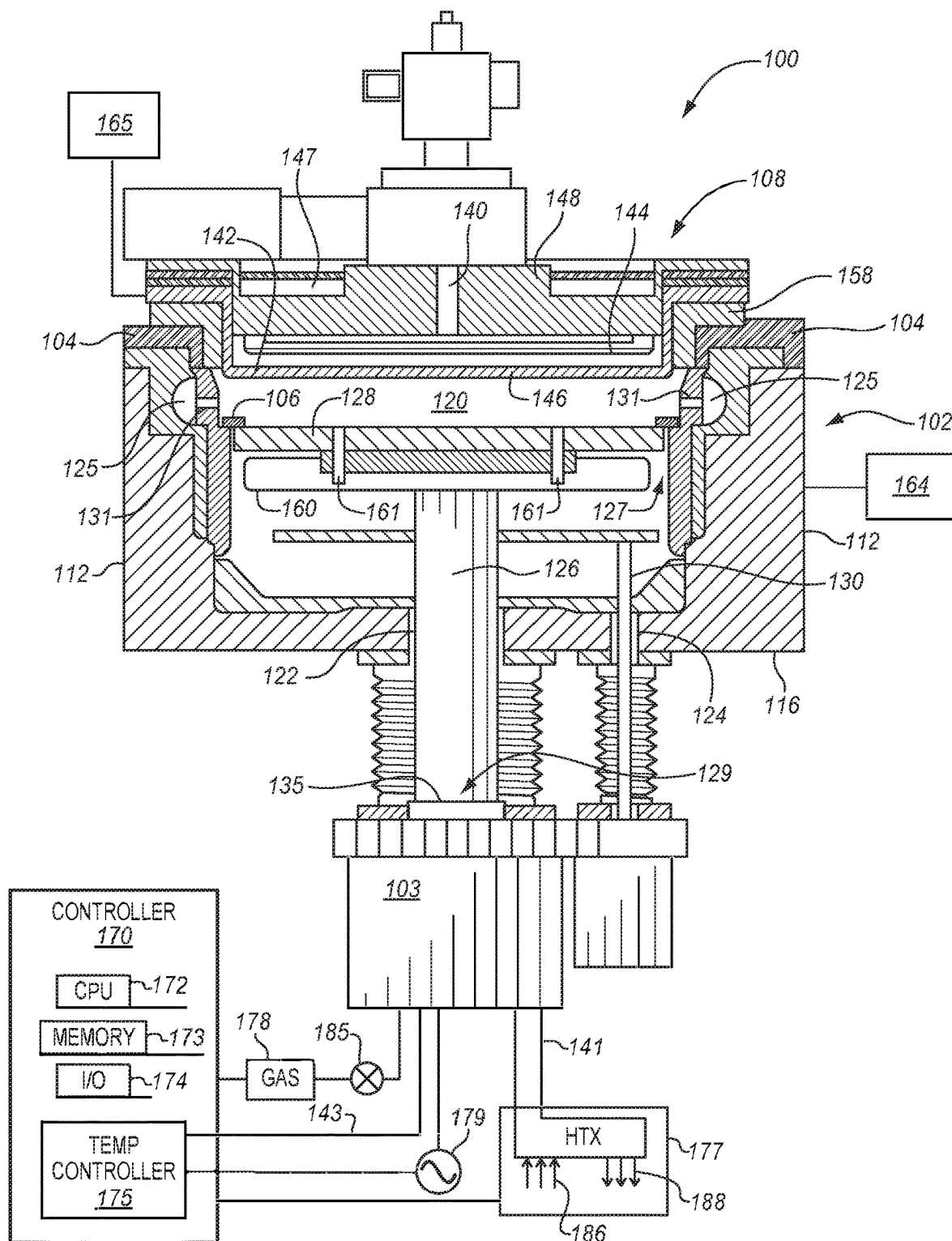


FIG. 9



1

SUBSTRATE CARRIER WITH ARRAY OF INDEPENDENTLY CONTROLLABLE HEATER ELEMENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 15/241,326, filed on Aug. 19, 2016, the entire contents of which are hereby incorporated by reference herein.

FIELD

The present description relates to the field of semiconductor and micromechanical substrate processing using a substrate carrier in a chamber and, in particular, independently controlling heating to different areas of the substrate using heating elements.

BACKGROUND

Semiconductor and micromechanical systems are formed on substrates, such as silicon wafers. A complex sequence of operations involving depositing, etching, shaping, patterning and washing are applied to the substrate to form the tiny structures that form the semiconductor and micromechanical parts on the substrate. These structures are made smaller and closer together in order to reduce cost. The smaller structures require less material, less power to operate, and less space to house. The smaller structures are also often faster than larger structures and may have many more advantages.

In order to make smaller structures, the fabrication processes must be more precise. Every aspect of the process is improved over time to enable smaller structures. Many of the fabrication processes, such as electron beam deposition, plasma deposition, plasma-enhanced chemical vapor deposition (PECVD), resist stripping, and plasma etching, etc. are affected by the temperature of the silicon wafer. If the temperature of the wafer in one location is different from that in another location, then the result of the process will be different in the different locations. In addition, if the temperature is different from the temperature for which the process was designed, then the results of the process will not be what was designed. As a result, temperature variations during fabrication may cause some structures to work poorly or even be inoperable.

The temperature of a substrate in a processing chamber can be measured using an infrared camera or a heat sensor on the substrate. In some cases a special wafer is used that has one or more thermal sensors and that store temperatures in a memory during a test process. A process is performed with the special wafer in the chamber and then the process is adjusted based on the measured temperatures.

SUMMARY

A substrate carrier is described with an array of independently controllable heater elements. In one example an apparatus includes a substrate carrier to carry a substrate for processing, a plurality of resistive heating elements in the carrier to heat the substrate by heating the carrier, a power supply to supply power to the heating elements, a power controller to provide a control signal, the control signal to control an amount of current applied to each of the heating elements, and a plurality of power interfaces in the carrier each coupled to a heating element to receive the power from

2

the power supply and the control signal from the controller and to modulate the power applied to a respective coupled heating element in response to the control signal.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the present invention are illustrated by way of example, and not limitation, in the figures of the accompanying drawings in which:

FIG. 1 is a process flow diagram of a subtractive approach to determining a temperature profile of a substrate attached to a substrate carrier according to an embodiment.

FIG. 2 is a process flow diagram of an additive approach to determining a temperature profile of a substrate attached to a substrate carrier according to an embodiment.

FIG. 3 is a process flow diagram of an additive and subtractive approach to determining a temperature profile of a substrate attached to a substrate carrier according to an embodiment.

FIG. 4 is a process flow diagram of a passive carrier approach to determining a temperature profile of a substrate attached to a substrate carrier according to an embodiment.

FIG. 5 is a graph of resistance over temperature of an electrical response characteristic of a thermal device and a heating element according to an embodiment.

FIG. 6 is a graph of current over time of power from a carrier power supply according to an embodiment.

FIG. 7 is a diagram of a control box coupled to a heated substrate carrier according to an embodiment.

FIG. 8 is an isometric view of an electrostatic chuck in accordance with an embodiment of the invention.

FIG. 9 is a schematic of a plasma etch system including a chuck assembly in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

As described herein an array of heating elements in a support, carrier, pedestal, or chuck that carries a substrate in a processing chamber may be used to measure the temperature of the substrate support. The heating elements have a resistance that is related to temperature so that the resistance of a heating element can be measured as an indication of the temperature of the support. This provides an indication of the temperature of the substrate immediately above the support. The array of heater elements provides independent temperature measurements at different positions on the support. The different measurements allow temperature variations across the substrate to be measured. This allows for the heaters to be operated to even out the temperature or for the support to be modified to correct for the inconsistent temperatures.

In some cases, an electrostatic chuck (ESC) is fitted with an array of heater elements to allow for the temperature of the ESC to be adjusted differently at different positions across the surface of the ESC. While the present description is presented in the context of an ESC, the structures and techniques may also be applied to other types of substrate carriers. The array of heater elements may also be used as temperature sensors or thermal sensors. This allows for the temperature of the substrate to be determined. The corresponding heater element may then be activated or deactivated to accommodate the measured temperature.

The heater elements are wired to receive an electrical current and the heating elements by nature typically have a linear relationship of resistance to temperature. The heater elements that are often used in an ESC have a metallization

3

material formulated of tungsten and alumina. This metalized material like many others responds to temperature with a linear relationship that can be used for measurement and control.

By using the heater elements, external temperature sensors are avoided, simplifying the ESC and the processing chamber. There may also be more accurate information on temperature variations across the substrate by measuring the temperature at many closely arrayed points.

The thermal sensor data from the heater elements can be used to feed open loop models or a time based closed loop control PID (Proportional-Integral-Derivative) scheme. The measurements may be taken inside and outside the RF (Radio Frequency) hot environment and may be implemented in many different ways in plasma etch and other processes. The thermal sensor may be used for open loop verification of the ESC and substrate temperature, for closed loop control of the ESC and substrate temperature, and for diagnostics of damaged heating arrays or cooling channel elements.

An alternative to the described approach is to place a thermal sensor near each heater element. In such a case, each heater element requires a pair of wires to control the rate at which it applies heat to the ESC. Each thermal sensor also requires a pair of wires to send temperature readings. The footprint for this system is larger to accommodate the wiring, the switching, the power distribution, and the control circuitry of the thermal probes or RTDs (Resistance Temperature Detector)

The larger footprint takes space which could be used for switching elements or other logic devices in the ESC. There would also be space in a control box to receive all the additional wires and track and maintain all of the temperature measurements. In addition to the space required, complicated I/O and mechanical interfaces to external processing devices are required. This additional cost and complexity might lead to designs that are constrained to fewer less accurate measurements in order to reduce cost.

The described techniques may be applied to wafers, pedestals, carriers, and electrostatic chucks with multiple heater zones. These may also include bulk heating zones. In some embodiments there are more than 150 mini resistive heaters that are used in the RF hot environment, but there may be 300 or more. This may all be accommodated with a 300 mm silicon wafer carrier. As described below, the control architecture may be scaled to support the hundreds of heating zones. This architecture provides real time control to devices both internally and externally of the RF hot environment, but does not require extra temperature measurement hardware.

To measure the temperature, the heating current to a particular heater element is turned off. A measurement current is then driven through the heater element. Sensor hardware may be used to measure the full current or a partial current going through the heater element. In some embodiments a PWM (Pulse Width Modulation) frequency may be used.

The relationship between the measured current and the temperature of the ESC can be accurately measured in several different ways. One way is by timing the activated array element and measuring the current and the voltage used by the element. Current or voltage or both may be measured by devices such as inline shunt resistors, current clamps, Hall Effect sensors, and voltage dividers. The signal can be digitized by analog to digital converters for later processing.

4

FIG. 1 is a process flow diagram of a subtractive approach to determining a temperature profile of a substrate attached to an electrostatic chuck or any other substrate carrier featuring heaters as described herein. At 2 a first combined current load is measured. This current load is the current consumed by all of the heating elements of the carrier or of a group of heating elements that are all powered with a single current supply. In this process, single heating elements or small groups of heating elements may be individually turned on or off.

At 4 the power is disconnected from one of the heating elements of the carrier. In this example, the heating elements have a common power supply but may individually be controlled or controlled in groups. Each heating element or group of heating elements may be switched off, de-powered, or disconnected in some other way so that the selected heating element is no longer consuming power and is no longer heating the carrier or the substrate that is being carried. This is referred to as element i , where i is a variable that is incremented through all of the heating elements that will be measured.

With the selected heating element disconnected the current load is measured again at 6. This provides a second combined current load for all heating elements after disconnecting power to the first selected heating element. At 8 the difference between the first and the second combined current loads is determined. This provides the current that was being consumed by the selected heating element or element i . At 10 the temperature of the selected heating element is determined.

The temperature of the heating element is directly related to its resistance. The resistance may be determined using the current and known voltage. If the voltage is not known then this may also be measured at the same time as the current load. The resistance has a linear relationship to the temperature of the heating element which is the same as the temperature of the carrier in the area of the heating element. The temperature of the heating element may then be related to the substrate that is being carried by using a model or a look up table based on empirical measurements. With the temperature at the heating element determined, the system may then move to reconnect power to the selected heating element at 12.

At 14 another heating element is selected. If the first element is designated as element 1, wherein $i=1$, then i is incremented and the next element is element 2, wherein $i=2$. After incrementing i or after simply selecting the next heating element, the process returns to 2 and the operations of disconnecting, measuring a current load, determining a difference, and temperature and reconnecting the heater element are repeated for each of the other heating elements of the to determine a temperature at each of the heating elements of the carrier.

After all of the heating element are measured then the process ends. In this process, the selection of heating elements may be ordered with an offset or spacing. Since a heating element is temporarily turned off to make the temperature measurement, the carrier will be slightly cooler in the area of that heating element. The next heating element to be measured may be spaced some distance from the last heating element so that the local cooling does not affect the next measurement. The carrier will have time to recover before another heating element in the same area is measured.

As mentioned above inside of measuring the temperature at each heating element, the heating elements may be grouped. In this case, the temperature will be averaged over the group of heating elements. As an example, the deter-

5

mined current, voltage, or resistance may be divided or distributed among the heating elements of the measured subset. This may be done simply by dividing by the number of heating elements or in another way. The temperature may then be determined using the divided resistance.

FIG. 2 is a process flow diagram of an additive approach to determining a temperature profile of a substrate attached to an electrostatic chuck or other substrate carrier with heaters. At 22 a first one of the heating elements is switched off, de-powered, or disconnected from using power in some way. At 24 a first combined current load is measured. This current load is the current consumed by all of the heating elements of the carrier or of a group of heating elements that are all powered with a single current supply excluding the heating element for which the temperature is to be measured. In this process, a single heating element or small group of heating elements turned off. Power is supplied to the other heating elements and this is what is measured. At 26 power is restored to the selected heating element i , and at 28 the current is measured again. As in the example of FIG. 1, the voltage may also be measured with the current or the voltage may be maintained at a constant depending on the particular power supply implementation.

At 30 the difference between the two measurements is determined. This isolates the power consumed by the selected heating element, i . At 32 this difference is used to determine a temperature of the heating element. The temperature is directly related to the resistance which may be determined using the current and the voltage difference. This temperature determination may then be stored in a log with a measurement time for use in modifying the process, detecting production variations, improving yield and other purposes.

At 34 the process ends if all of the heating elements have been measured or it returns to the beginning to go to the next heating element. This may be performed by incrementing i to represent the next heating element and then repeating the operations of disconnecting power, measuring the current, connecting the power, measuring the current again, determining a difference, and determining a temperature for each of the other heating elements.

FIG. 3 is a process flow diagram of an alternative additive and subtractive approach to determining a temperature profile of a substrate attached to an electrostatic chuck or any other substrate carrier featuring heaters as described herein. As above, this process is particularly well suited to use during a plasma process but may also be used in other circumstances. At 40 the voltage supplied to all of the resistive heating elements is measured. The measurement may be made at a central power supply that supplies all of the resistive heating elements together. If the heating elements are powered in smaller groups, e.g. four separate power supplies are each used for a separate one fourth of the heating elements, then this process may be applied separately to each group. This operation may also be applied to the first two processes described above.

At 42 a first combined current load is measured. This current load is the current consumed by all of the heating elements of the carrier or of a group of heating elements that are all powered with a single current supply. In this process, single heating elements or small groups of heating elements may be individually turned on or off. This process is performed while the heating elements are in normal operation. Accordingly, most of the heating elements will be in an intermediate power state as neither fully on nor fully off. In other words the current duty cycle will be at some intermediate value.

6

For the next operation a particular single one of the heating elements is selected as the start of the process. This element is designated e.g. as element 1 or element 0 and a variable $i=0$ to track the process through each element. At 44 the power is increased from the first heating element $i=0$. In some embodiments, the heating element goes to full ON or a 100% duty cycle. This ON states is required only long enough to measure the current. At 46, the current load of all of the heating elements is measured again after increasing the power. At 48 the power to the single heating element is then switched to OFF. The other heating elements of the carrier are not affected.

With the selected heating element disconnected at 48 the current load is measured again at 50. This provides a third combined current load for all heating elements after disconnecting power to the first selected heating element. The carrier may then be returned to normal operation and the temperature of the carrier may be stabilized at 52.

At 56 the differences between the first, second and third combined current loads are determined. This provides the current that was being consumed by the selected heating element or element i . At 56 the temperature of the selected heating element is determined using these differences.

At 58 another heating element is selected. If the first element is designated as element 1, wherein $i=1$, then i is incremented and the next element is element 2, wherein $i=2$. After incrementing i or after simply selecting the next heating element, the process returns to 40 or to 42 if the voltage is fixed and the operations are repeated at each of the heating elements of the carrier. After all of the heating element are measured then the process ends.

FIG. 4 is a process flow diagram of an alternative approach to determining a temperature profile of a substrate attached to a passive electrostatic chuck or any other substrate carrier featuring heaters as described herein. This process is particularly well suited to use as a test wafer during a plasma process but may also be used in other circumstances. To instrument processes in a particular chamber, a test wafer is often used. The test wafer has a top surface with a hundred or more thermal sensors coupled to memory cells. The test wafer is attached a carrier while a process is being run and the test wafer logs temperature at each thermal sensor during the process. After the process, the test wafer is removed and the temperature logs are read. The process may then be adjusted based on the temperature data.

The process of FIG. 4 collects similar data using a heated carrier with a passive wafer on top. The heaters are used to collect the temperature data without the need for a test wafer. The temperature of the wafer may be estimated based on the temperature of the carrier immediately below the carrier. For such a test, the heating elements are not used for heating but only for measuring temperature. In contrast to the test wafer, the process of FIG. 4 allows temperature data to be collected in real time or during the process rather than waiting until the end of the process. This allows the test process to be modified during the test to obtain the desired results more quickly.

At 60 the voltage supplied to all of the resistive heating elements is measured. This is normally the primary power supply voltage that is supplied to all of the resistive heating elements. The current at 62 will be zero or negligible because the carrier is not being used to add heat to the test process. At 64 a particular single one i of the heating elements is selected as the start of the process and power is applied. This will be a known power selected to provide the best temperature measurement. As explained below a higher

7

current will typically provide a higher response to temperature. With the one powered heating element, the current load is measured at **66**. This current is almost completely due to the selected heating element. The selected heating element may then be returned to OFF, the normal state for this cycle.

At **68** the difference between the first and second combined current loads is determined. This subtracts out noise, current leakage and other factors that move the OFF state current away from zero. At **70** the temperature of the selected heating element is determined using these differences.

At **72** another heating element is selected and the process is repeated until all of the heating elements are measured. That is the end of this measurement cycle. As with FIGS. **1-3**, the process may be completed through all of the heating elements for as long as desired. This provides a continuous temperature map over time.

FIG. **5** is a graph of an electrical response characteristic of a typical commercial resistive thermal device and of a typical commercial heating element for comparison. The vertical scale shows the resistance of the component and the horizontal scale shows the temperature of the component. The upper line corresponds to a resistive thermal device designed specifically for measuring temperature by changing resistance. As shown at room temperature of about 20° C., the resistance is about 108Ω. This increases to almost 140Ω at 100° C. As shown, the response of the thermal device within the illustrated temperature range is approximately linear. This allows temperatures to be easily and quickly determined based on the measured resistance.

The response of a heating element is shown on the lower line. This line is also approximately linear but with a lower slope. As a result the temperature measurement is less accurate. This is a result of the thermal device being specifically designed to exhibit a large change in response to temperature. The heating element is designed to efficiently convert current to heat. With modifications, the response of the heating element may be improved. In this example, at room temperature the heating element has a resistance of about 70Ω. At 100° C., the resistance has increased to about 90Ω. While the thermal device shows a change of about 30Ω between the two temperatures, the heating element shows a change of only about 20Ω through the same temperature.

FIG. **6** is a graph of current over time from a substrate carrier power supply as heating elements are turned on and off. This shows how the additive and subtractive techniques described above may be used. The graph shows different current levels from a common main power supply as the state of various heating elements is changed. The first state **230** corresponds to an average current or power when most of the heating elements are powered. At **232** one or more heating elements are switched from ON to OFF. This results in a reduction in the total current. At **234** the low current stage is ended and at **236** the heating elements are returned to an ON state. The difference between the current at **234** and at **236** may be used to determine the current consumed by the heating elements that were turned from ON to OFF. This may then be used to determine the temperature in the vicinity of those heating elements.

In a similar way the current at **238** reflects an average or normal power and at **240** a second set of one or more heating elements are switched from an OFF to an ON state. This increases the total current. At **242** the ON cycle ends and at **244** the heating elements are returned to an OFF state. The difference between the current at **242** and **244** may be used to determine the current draw and with it the temperature in the vicinity of the second set of heating elements. The shape

8

of curve is shown as rectangular in order to simplify the drawing figure. In a real system there will be settling time after each change as the heating elements and the power supply adjust to the change in state. The response of the heating elements and of the power supply will not be immediate and direct as shown.

FIG. **7** is a block diagram of a substrate carrier and a temperature control system suitable for use with the method described above. The system has a control box **502** that is coupled to a terminal **504**. The terminal may be in the form of a conventional computer that runs process control or temperature control software with a user interface to allow an operator to control the machine processes. The terminal may have a processor coupled to a mass storage medium, a user interface and an interface to the control box. The terminal may have other components such as high speed memory, wireless or wired communications interfaces, additional processors, etc. The mass storage may be in the form of a machine-readable medium having instructions, parameters, and various logs, in solid state, optical, or magnetic storage. The control box controls the operation of the substrate carrier **506** in response to instructions or commands from the terminal. The control box may be able to operate autonomously subject to general commands from the terminal. The control box may control other functions of the carrier such as clamp electrodes, coolants, gas injection, and other functions not shown here in order not to obscure the features of temperature measurement shown here.

The control box includes a heating element power controller **510** and a power supply **512**. The power supply provides a single power feed **516** on a single line to a fan-out distributor **518** within the substrate carrier **506**. The fan-out distributor supplies the power from the power supply **512** to all of the heating elements **530** of the carrier **506**. This power may be significant depending on the implementation. In the illustrated embodiment there are 150 heating elements each capable of drawing 10 watts, so the power supply provides 1500 watts to the carrier. The voltage, current, and other parameters are measured and controlled by the power supply **512** in the control box.

The power controller of the control box sends control signals through a data interface **514** to a carrier controller **522** of the carrier **506**. The control signals may be used to adjust the power supplied to each of the heating elements **530**, change from an ON state to an OFF state and to adjust other parameters of the carrier. In this example, the data signals set the parameters of operation of each of the heaters based on information from the terminal or generated within the control box power controller. The control signal **514** may be a sequence of packets where each packet sets parameters for a different one of the heating elements. Packet headers or identification fields might be used to identify a particular one of the heating elements so that the status of a particular heating element may be changed with any packet and before it might have a turn in an ordered sequence of packets.

In this example, the carrier controller receives the status update packets for each heating element and generates a different PWM (Pulse Width Modulated) analog signal to each of the heating elements. To set a particular heater to an OFF state, the PWM signal is a zero duty cycle signal or a low signal. To set different amounts of heating, the controller adjusts the duty cycle of the respective PWM signal for each heating element. This allows each heating element to be independently controlled using a unique signal to that element. In this example there are 150 unique individual PWM connections, one for each heating element. The PWM signals are generated by the carrier controller and received at a

power interface for each heating element. The unique signals from the carrier controller may be very simple so that they are only an on/off cycle of an optical signal indicating the duty cycle of the PWM pulses. These may be converted directly into a gate driver of an amplifier to the respective heating element.

At each respective heating element **530**, there is a power interface **528** and the actual resistive heating element **530**. The power interface receives the PWM signal at an isolator **532** such as an opto-isolator and applies the PWM signal to an amplifier. The amplifier receives power **520** from the fan-out distributor and modulates the received power based on the PWM signal. The modulated power is applied to the resistive heater **530** to heat the carrier and therefore to heat the substrate. The optical signal is used within the carrier to shield the sensitive signal from noise within the carrier. The plasma, bias power, active ions and other aspects of a plasma chamber may interfere with an electrical signal and especially an analog electrical signal.

Using this system, the terminal **504** may be used to initiate a temperature measurement cycle. Alternatively, the control box **510** may operate a temperature measurement cycle. The control box will send control signals **514** to the carrier controller **522** to set the duty cycles of particular ones of the heating elements to zero and then back again to normal in a particular order and with a particular timing as described in FIGS. 1-4. The power supply will monitor the changes in current draw from the heaters as each heater is turned on and off throughout the carrier. This temperature information may be used to increase or decrease the duty cycle applied to each heater or to provide metrics on a particular process. Each heating element may be switched from ON or OFF in microseconds and a change in heating requires a few seconds to cause a new stable temperature at the carrier. For a process that takes a few minutes, the temperature may be measured many times through the process.

Using 150 heaters provides very specific information about a small area of the carrier. More or fewer heaters may be used, depending on the desired precision. In addition, since each heating element is individually controlled, the controller may test the temperature only at some of the heaters. Instead of 150 different temperature locations, the controller may use only 50 different locations by measuring the temperature at only 50 of the available heaters. While 150 heating elements are suggested, there may be more or fewer depending on the desired precision. There may also be other heaters such as a few larger high power heaters to provide more heating with a coarse level of control. The coarse level of control may then be adjusted using the many smaller heaters. While in this example each heater is controlled individually, several heaters may be connected to a single PWM input and amplifier **528** so that a group of heaters are controlled as a group.

FIG. 8 is an isometric view of an assembled electrostatic chuck. A support shaft **212** supports a base plate **210** through an isolator **216**. A middle isolator plate **208** and an upper cooling plate **206** are carried by the base plate. The top cooling plate **206** carries a dielectric puck **205** on the top surface of the heater plate. The puck has an upper circular platform to support a workpiece **204** and a lower concentric circular base **207** to attach to the heater plate. The upper platform has internal electrodes to electrostatically attach the workpiece. The workpiece may alternately be clamped, vacuumed or attached in another way.

There is an adhesive bond **218** between the puck **215** and the top cooling plate **206** to hold the ceramic of the top plate to the metal of the cooling plate. Heaters may be formed in

the top plate or the middle heater plate. In such an embodiment, the middle plate performs other functions but is no longer the location of the heaters. The carrier controller may be attached to the cooling plate or to any other location. The resistive heating elements and associated power interface may be embedded into the ceramic of the puck. This places the heaters as close as possible to the substrate on the top plate for the greatest effect.

The ESC is able to control the temperature of the workpiece using resistive heaters in the puck as described above. In addition, coolant fluid may be used in the cooling plate. Electrical power, control signals, coolant, gases, etc. are supplied to the coolant plate **206** and the puck **205** through the support shaft **212**. The ESC may also be manipulated and held in place using the support shaft.

FIG. 9 is a partial cross sectional view of a plasma system **100** having a pedestal **128** according to embodiments described herein. The pedestal **128** has an active cooling system which allows for active control of the temperature of a substrate positioned on the pedestal over a wide temperature range while the substrate is subjected to numerous process and chamber conditions. The plasma system **100** includes a processing chamber body **102** having sidewalls **112** and a bottom wall **116** defining a processing region **120**.

A pedestal, carrier, chuck or ESC **128** is disposed in the processing region **120** through a passage **122** formed in the bottom wall **116** in the system **100**. The pedestal **128** is adapted to support a substrate (not shown) on its upper surface. The substrate may be any of a variety of different workpieces for the processing applied by the chamber **100** made of any of a variety of different materials. The pedestal **128** may optionally include heating elements (not shown), for example resistive elements, to heat and control the substrate temperature at a desired process temperature. Alternatively, the pedestal **128** may be heated by a remote heating element, such as a lamp assembly.

The pedestal **128** is coupled by a shaft **126** to a power outlet or power box **103**, which may include a drive system that controls the elevation and movement of the pedestal **128** within the processing region **120**. The shaft **126** also contains electrical power interfaces to provide electrical power to the pedestal **128**. The power box **103** also includes interfaces for electrical power and temperature indicators, such as a thermocouple interface. The shaft **126** also includes a base assembly **129** adapted to detachably couple to the power box **103**. A circumferential ring **135** is shown above the power box **103**. In one embodiment, the circumferential ring **135** is a shoulder adapted as a mechanical stop or land configured to provide a mechanical interface between the base assembly **129** and the upper surface of the power box **103**.

A rod **130** is disposed through a passage **124** formed in the bottom wall **116** and is used to activate substrate lift pins **161** disposed through the pedestal **128**. The substrate lift pins **161** lift the workpiece off the pedestal top surface to allow the workpiece to be removed and taken in and out of the chamber, typically using a robot (not shown) through a substrate transfer port **160**.

A chamber lid **104** is coupled to a top portion of the chamber body **102**. The lid **104** accommodates one or more gas distribution systems **108** coupled thereto. The gas distribution system **108** includes a gas inlet passage **140** which delivers reactant and cleaning gases through a showerhead assembly **142** into the processing region **120B**. The showerhead assembly **142** includes an annular base plate **148** having a blocker plate **144** disposed intermediate to a faceplate **146**.

11

A radio frequency (RF) source **165** is coupled to the showerhead assembly **142**. The RF source **165** powers the showerhead assembly **142** to facilitate generation of plasma between the faceplate **146** of the showerhead assembly **142** and the heated pedestal **128**. In one embodiment, the RF source **165** may be a high frequency radio frequency (HFRF) power source, such as a 13.56 MHz RF generator. In another embodiment, RF source **165** may include a HFRF power source and a low frequency radio frequency (LFRF) power source, such as a 300 kHz RF generator. Alternatively, the RF source may be coupled to other portions of the processing chamber body **102**, such as the pedestal **128**, to facilitate plasma generation. A dielectric isolator **158** is disposed between the lid **104** and showerhead assembly **142** to prevent conducting RF power to the lid **104**. A shadow ring **106** may be disposed on the periphery of the pedestal **128** that engages the substrate at a desired elevation of the pedestal **128**.

Optionally, a cooling channel **147** is formed in the annular base plate **148** of the gas distribution system **108** to cool the annular base plate **148** during operation. A heat transfer fluid, such as water, ethylene glycol, a gas, or the like, may be circulated through the cooling channel **147** such that the base plate **148** is maintained at a predefined temperature.

A chamber liner assembly **127** is disposed within the processing region **120** in very close proximity to the sidewalls **101**, **112** of the chamber body **102** to prevent exposure of the sidewalls **101**, **112** to the processing environment within the processing region **120**. The liner assembly **127** includes a circumferential pumping cavity **125** that is coupled to a pumping system **164** configured to exhaust gases and byproducts from the processing region **120** and control the pressure within the processing region **120**. A plurality of exhaust ports **131** may be formed on the chamber liner assembly **127**. The exhaust ports **131** are configured to allow the flow of gases from the processing region **120** to the circumferential pumping cavity **125** in a manner that promotes processing within the system **100**.

A system controller **170** is coupled to a variety of different systems to control a fabrication process in the chamber. The controller **170** may include a temperature controller **175** to execute temperature control algorithms (e.g., temperature feedback control) and may be either software or hardware or a combination of both software and hardware. The system controller **170** also includes a central processing unit **172**, memory **173** and input/output interface **174**. The temperature controller receives a temperature reading **143** from a sensor (not shown) on the pedestal. The temperature sensor may be proximate a coolant channel, proximate the wafer, or placed in the dielectric material of the pedestal. The temperature controller **175** uses the sensed temperature or temperatures to output control signals affecting the rate of heat transfer between the pedestal assembly **142** and a heat source and/or heat sink external to the plasma chamber **105**, such as a heat exchanger **177**.

The system may also include a controlled heat transfer fluid loop **141** with flow controlled based on the temperature feedback loop. In the example embodiment, the temperature controller **175** is coupled to a heat exchanger (HTX)/chiller **177**. Heat transfer fluid flows through a valve (not shown) at a rate controlled by the valve through the heat transfer fluid loop **141**. The valve may be incorporated into the heat exchanger or into a pump inside or outside of the heat exchanger to control the flow rate of the thermal fluid. The heat transfer fluid flows through conduits in the pedestal assembly **142** and then returns to the HTX **177**. The temperature of the heat transfer fluid is increased or decreased

12

by the HTX and then the fluid is returned through the loop back to the pedestal assembly.

The HTX includes a heater **186** to heat the heat transfer fluid and thereby heat the substrate. The heater may be formed using resistive coils around a pipe within the heat exchanger or with a heat exchanger in which a heated fluid conducts heat through an exchanger to a conduit containing the thermal fluid. The HTX also includes a cooler **188** which draws heat from the thermal fluid. This may be done using a radiator to dump heat into the ambient air or into a coolant fluid or in any of a variety of other ways. The heater and the cooler may be combined so that a temperature controlled fluid is first heated or cooled and then the heat of the control fluid is exchanged with that of the thermal fluid in the heat transfer fluid loop.

The valve (or other flow control devices) between the HTX **177** and fluid conduits in the pedestal assembly **142** may be controlled by the temperature controller **175** to control a rate of flow of the heat transfer fluid to the fluid loop. The temperature controller **175**, the temperature sensor, and the valve may be combined in order to simplify construction and operation. In embodiments, the heat exchanger senses the temperature of the heat transfer fluid after it returns from the fluid conduit and either heats or cools the heat transfer fluid based on the temperature of the fluid and the desired temperature for the operational state of the chamber **102**.

Electric heaters (not shown) are used in the ESC to apply heat to the workpiece assembly. The electric heaters in the form of resistive elements are coupled to a power supply **179** that is controlled by the temperature control system **175** to energize the heater elements to obtain a desired temperature.

The heat transfer fluid may be a liquid, such as, but not limited to deionized water/ethylene glycol, a fluorinated coolant such as Fluorinert® from 3M or Galden® from Solvay Solexis, Inc. or any other suitable dielectric fluid such as those containing perfluorinated inert polyethers. While the present description describes the pedestal in the context of a PECVD processing chamber, the pedestal described herein may be used in a variety of different chambers and for a variety of different processes.

A backside gas source **178** such as a pressurized gas supply or a pump and gas reservoir are coupled to the chuck assembly **142** through a mass flow meter **185** or other type of valve. The backside gas may be helium, argon, or any gas that provides heat convection between the wafer and the puck without affecting the processes of the chamber. The gas source pumps gas through a gas outlet of the pedestal assembly described in more detail below to the back side of the wafer under the control of the system controller **170** to which the system is connected.

The processing system **100** may also include other systems, not specifically shown in FIG. 4, such as plasma sources, vacuum pump systems, access doors, micromachining, laser systems, and automated handling systems, inter alia. The illustrated chamber is provided as an example and any of a variety of other chambers may be used with the present invention, depending on the nature of the workpiece and desired processes. The described pedestal and thermal fluid control system may be adapted for use with different physical chambers and processes.

As used in the description of the invention and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and

encompasses any and all possible combinations of one or more of the associated listed items.

The terms “coupled” and “connected,” along with their derivatives, may be used herein to describe functional or structural relationships between components. It should be understood that these terms are not intended as synonyms for each other. Rather, in particular embodiments, “connected” may be used to indicate that two or more elements are in direct physical, optical, or electrical contact with each other. “Coupled” may be used to indicate that two or more elements are in either direct or indirect (with other intervening elements between them) physical, optical, or electrical contact with each other, and/or that the two or more elements co-operate or interact with each other (e.g., as in a cause an effect relationship).

The terms “over,” “under,” “between,” and “on” as used herein refer to a relative position of one component or material layer with respect to other components or layers where such physical relationships are noteworthy. For example in the context of material layers, one layer disposed over or under another layer may be directly in contact with the other layer or may have one or more intervening layers. Moreover, one layer disposed between two layers may be directly in contact with the two layers or may have one or more intervening layers. In contrast, a first layer “on” a second layer is in direct contact with that second layer. Similar distinctions are to be made in the context of component assemblies.

It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, while flow diagrams in the figures show a particular order of operations performed by certain embodiments of the invention, it should be understood that such order is not required (e.g., alternative embodiments may perform the operations in a different order, combine certain operations, overlap certain operations, etc.). Furthermore, many other embodiments will be apparent to those of skill in the art upon reading and understanding the above description. Although the present invention has been described with reference to specific exemplary embodiments, it will be recognized that the invention is not limited to the embodiments described, but can be practiced with modification and alteration within the spirit and scope of the appended claims. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

What is claimed is:

1. A method comprising:

supplying power to a plurality of heating elements from a common power supply, the heating elements being in a substrate support to heat a substrate by heating the substrate support, the substrate support to carry the substrate during processing;

generating a control signal from a power controller to control an amount of current applied to each of the heating elements;

receiving the power from the power supply at each of a plurality of power interfaces in the substrate support each coupled to a heating element;

receiving the control signal at each of the plurality of power interfaces, the receiving comprising receiving the control signal from the power controller at a carrier controller embedded in the substrate support, wherein the control signal is a serial packetized control signal and generating a unique control signal to each heating interface from the carrier controller; and

modulating the power applied to a respective coupled heating element by a respective power interface in response to the control signal.

2. The method of claim 1, further comprising receiving instructions at the power controller from a program operating on a terminal to change a power modulation to a heating element of the plurality of heating elements and changing the control signal at the power controller in response to a received instruction.

3. The method of claim 1, wherein the control signal from the power controller is an optical signal.

4. The method of claim 1, further comprising:

measuring a voltage of the common power supply;

measuring a current of the common power supply;

generating a control signal to change a power state of a selected one of the heating elements;

measuring a current of the common power supply after changing the power state;

determining a difference of the first and second current measurement; and

determining a temperature of the selected one of the heating elements using the determined current measurement difference.

5. The method of claim 1, wherein the substrate support is a chuck.

6. The method of claim 5, wherein the chuck is an electrostatic chuck.

* * * * *